

Whisper small.en - A113X end-to-end review

One public 11-second speech sample. From-scratch C runtime. Greedy English transcription. No timestamps.

Median duration 45.047 s	Median RTF 4.0952	Cumulative gain 13.08x	Median-run RSS 251,396 KiB
Decoder throughput 3.834 tok/s	Overall throughput 0.555 tok/s	Smoke-case WER 0 / 22	Process swap 0

Input and output

Field	Exact value
Input	whisper.cpp samples/jfk.wav
Input SHA-256	59dfb9a4acb36fe2a2affc14bacbee2920ff435cb13cc314a08c13f66ba7860e
Format	352,078-byte mono PCM16 WAV, 16,000 Hz, 11.000 seconds
Model input	Compact: 1,100 Mel frames and 550 encoder frames
Decode policy	English transcription, batch one, greedy, no timestamps, maximum 64 steps
Output	25 text tokens; EOT on decode step 26

Exact emitted text, including its leading space:

And so my fellow Americans, ask not what your country can do for you, ask what you can do for your country.

Reference: And so, my fellow Americans, ask not what your country can do for you, ask what you can do for your country.

Normalization: lowercase ASCII words; remove punctuation and leading whitespace. Result: 0 substitutions, 0 insertions and 0 deletions over 22 reference words. This is one smoke case, not the relative-WER suite.

Benchmark

Run	Total	RTF	CPU	Peak RSS	Swap
E9 row reuse 1	45.810786 s	4.164617	380%	251,276 KiB	0
E9 row reuse 2	44.872162 s	4.079287	388%	251,488 KiB	0
E9 row reuse 3 / median	45.046989 s	4.095181	388%	251,396 KiB	0

Mean 45.243312 s; sample standard deviation 0.499160 s; coefficient of variation 1.103%.

Incremental result

Stage	Cumulative implementation	Duration	RTF	Increment	Cumulative
E1	fixed30, A3 encoder + D1 decoder	589.055851 s	53.5505	1.00x	1.00x
E2	E1 + compact input window	160.802747 s	14.6184	3.66x	3.66x
E3	E2 + Q4 output-column cache block	73.774966 s	6.7068	2.18x	7.98x
E4-E7	select 8 / 16 / 32 / 64-column block	61.882 to 52.207 s	5.6257 to 4.7461	1.19-1.03x	11.28x
E8	E7 + fixed 4-output NEON dot	48.214580 s median	4.3831	1.08x	12.22x
E9	E8 + 2-row x 4-output Q4 micro-kernel	45.046989 s median	4.0952	1.07x	13.08x

E1-E3, E8 and E9 are committed feature boundaries. E4-E7 are tile-selection experiments; exact intermediate source was not retained as a release commit.

Median-run path duration

Path	Duration	Share
Waveform to log-Mel	0.079188 s	0.18%
Stem + 12-layer encoder	38.436764 s	85.33%
Cross cache + decoder core	6.094089 s	13.53%
Tied output head	0.426832 s	0.95%
Measured runtime overhead	0.010116 s	0.02%
Total internal timer	45.046989 s	100%

Canonical resource run	Measured
Role	separate resource audit; excluded from the median
User / system CPU	176.42 / 0.86 s
Average CPU	388%
Maximum resident set	251,332 KiB
Process swaps	0
Device SwapFree before / after	1,005,484 / 1,005,484 KiB
Exit status	0

Exact command

```
OMP_NUM_THREADS=4 OMP_DYNAMIC=FALSE /usr/bin/time -v ./build/whisper-small-transcribe-a113x small.en-full-mixed.whenc
/root/whisper-bench-v1.8.0/whisper.cpp/samples/jfk.wav 64 compact
```

Verification

Verification is separate from benchmark duration. Quantized hidden-state error is measured but is not substituted for a WER acceptance test.

Boundary	Reference	Result
Real encoder stem	independent NumPy graph	RMSE 3.11536603e-8
Real encoder layer 0	same pinned mixed image	RMSE 0.0605037052
Real encoder final output	same pinned mixed image	RMSE 0.238280457
Mixed decoder, 3 steps	independent full-prefix decoder	next tokens match 3 / 3 on 1 and 4 threads
JFK transcription	public reference sentence	normalized word edits 0 / 22

Artifact identity

Artifact	Exact identity
Repository commit	51c629a
Full mixed image	214,878,912 bytes; SHA-256 8f34d629c5ec4b8a3af4c9de03252fd60ab08781538768c3a5171f6a74b82343
A113X binary SHA-256	1abaa6dc63a2b64b1f50fba30f91ebc25fdd92db83929d56af9c8ac203e1277
Q4 kernel SHA-256	fe7fe869451efa2bc6e4647accdfb555c83ea4be55955c4fbbd852429562142a
Target	Amlogic A113X; 4x Cortex-A53; max observed 1.416 GHz; 2,059,239,424 bytes RAM
Quantization	encoder Transformer matrices Q4; decoder and tied head Q8; other tensors F32

Limitations

Real-time transcription still fails at RTF 4.0952. Relative WER over a multi-file suite, timestamp output and long-audio window scheduling remain open. The three-run statistic covers one public smoke input only.

Raw evidence

Directory: `models/whisper-small.en/targets/a113x/benchmarks/jfk-11s/`

Machine-readable record: `row-reuse-result.json`

Files: `row-reuse-run-1.stdout.txt`, `row-reuse-run-1.time.txt`, `row-reuse-run-2.stdout.txt`, `row-reuse-run-2.time.txt`, `row-reuse-run-3.stdout.txt`, `row-reuse-run-3.time.txt`, `row-reuse-canonical.stdout.txt`, `row-reuse-canonical.time.txt`, `row-reuse-canonical.memory-before.txt`, `row-reuse-canonical.memory-after.txt`